

Stellarocq: Machine-Checked Certificates for Stellarator Equilibria

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Abstract

The output of a stellarator equilibrium code is a file of floating-point Fourier coefficients. Stellarocq states what such a file establishes as machine-checked theorems about the magnetic field it defines. The field is reconstructed from the exact numbers of a VMEC++ output file by VMEC's half-grid interpolation of the surface coefficients. The development proves that the residual it evaluates is $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ of that field, and that its Jacobian and metric differ from those VMEC forms from the same numbers by exact terms of second order in the radial step. Each claim about the field, a bound, a Fourier harmonic, a flux-surface integral or a Mercier term, is decided by a checker extracted from a Rocq proof that a passing verdict implies the claim. The certificates give the error of the half-grid rule on manufactured problems to a relative 10^{-12} , falling as h^2 away from the axis. On VMEC++ output they bound away from zero a harmonic of the force at a rational surface that radial refinement reduces far more slowly than the scheme's order, and they enclose the free-boundary pressure jump for every vacuum field between the answers of NESTOR and BIEST.

1 Introduction

The Variational Moments Equilibrium Code (VMEC) [1, 2] computes three-dimensional ideal-MHD equilibria with nested flux surfaces by minimizing the plasma energy over Fourier representations of the surfaces. It and its reimplementations VMEC++ [3, 4] sit at the centre of stellarator design and optimization. A run ends with a *wout* file: Fourier coefficients of the surfaces on a radial grid, profiles, and derived quantities such as the Mercier stability terms, all in binary64 floating point.

What such a file establishes about the field it describes is judged by convergence metrics internal to the solver and by diagnostics recomputed in floating point. Stellarocq states it as a theorem. The object of every statement is the field reconstructed from the exact numbers in the file by the solver's own interpolation of its surface coefficients, and the development proves how the Jacobian and metric of that field differ from the ones the solver forms. A claim about that field is decided by a checking function written in Rocq [5], proven to imply the claim when it returns **true**, and extracted to OCaml [6]; real-number semantics and interval arithmetic come from Coquelicot [7], Flocq [8] and CoqInterval [9].

Section 2 states the certified object and the theorem that identifies its residual with the ideal-MHD force, and Section 3 the certificate forms and the trust base. Section 4 reports what they establish and where each result stops: the error of the half-grid rule on manufactured problems, known to a relative 10^{-12} ; on VMEC++ output, a harmonic of the force on a rational surface, bounded away from zero over a box of states; the free-boundary pressure jump, enclosed for every vacuum field between the answers of two independent solvers; and the Mercier criterion of a coarse equilibrium, whose sign depends on how the radial derivatives are taken. The certified values enter VMEC++ as regression tests.

2 The certified object

Reconstruction. VMEC keeps the surface coefficients of R and Z on the full radial grid $s_j = jh$ and the stream function λ , the rotational transform ι and every field quantity on the half grid $s_{j+1/2}$. Stellarocq’s `Physics.v` writes the rule that connects them. On a half point s_h between nodes a and b , a coefficient of even poloidal mode number is the mean of its two node values and one of odd mode number is averaged with the $\sqrt{s}$ weight VMEC folds into its Jacobian,

$$c_k(s_h) = \frac{1}{2}(c_k(a) + c_k(b)) \quad (m_k \text{ even}), \quad c_k(s_h) = \frac{1}{2}\sqrt{s_h} \left(\frac{c_k(a)}{\sqrt{s_a}} + \frac{c_k(b)}{\sqrt{s_b}} \right) \quad (m_k \text{ odd}), \quad (1)$$

with radial derivatives the matching difference quotients and angular derivatives exact derivatives of the series $R = \sum_k c_k \cos(m_k u - n_k v)$, $Z = \sum_k z_k \sin(m_k u - n_k v)$. The half-grid λ and ι are the file’s own, Φ' is constant, and the pressure is the input’s profile. For statements at an arbitrary radius each coefficient of R and Z is the cubic Hermite interpolant of its half-point values and slopes, which agrees with the half-grid rule at the half points; λ follows the same parity rule linearly between half points, and ι is linear there.

Field and residual. With $\sqrt{g} = R(R_u Z_s - R_s Z_u)$, the field is $B^u = \Phi'(\iota - \lambda_u)/\sqrt{g}$, $B^v = \Phi'(1 + \lambda_u)/\sqrt{g}$, its covariant components are $B_i = g_{iu} B^u + g_{iv} B^v$, and $\mu_0 \sqrt{g} J^s = \partial_u B_v - \partial_v B_u$. The residual of $\mathbf{J} \times \mathbf{B} = \nabla p$ has the components

$$r_u = -\mu_0 \sqrt{g} J^s B^v, \quad r_v = \mu_0 \sqrt{g} J^s B^u, \quad r_s = (\partial_v B_s - \partial_s B_v) B^v - (\partial_s B_u - \partial_u B_s) B^u - \mu_0 \frac{dp}{ds}, \quad (2)$$

written in `Physics.v` as trees of a deep expression language. Read at a free radius, with every radial derivative that of the Hermite reconstruction, they form the continuum residual. Let $X(s, u, v) = (R \cos v, R \sin v, Z)$ be the reconstructed coordinate map, $\mathbf{e}_s$, $\mathbf{e}_u$ and $\mathbf{e}_v$ its partial derivatives, and $\mathbf{B} = B^u \mathbf{e}_u + B^v \mathbf{e}_v$ the reconstructed field as a function of (s, u, v) .

Theorem 1 (continuum_force_coords) *Let the slots of the residual’s expressions hold a state’s radii, coefficients, flux, transform and pressure derivative p' , and let (s_0, u_0, v_0) be a point where $\sqrt{g} \neq 0$. There let $\nabla s = \mathbf{e}_u \times \mathbf{e}_v / \sqrt{g}$, $\nabla u = \mathbf{e}_v \times \mathbf{e}_s / \sqrt{g}$ and $\nabla v = \mathbf{e}_s \times \mathbf{e}_u / \sqrt{g}$, and let $\partial_s \mathbf{B}$, $\partial_u \mathbf{B}$ and $\partial_v \mathbf{B}$ be the derivatives of $\mathbf{B}$ along the three coordinate lines through the point. Then the three outputs of the continuum residual there are $\mathbf{F} \cdot \mathbf{e}_s$, $\mathbf{F} \cdot \mathbf{e}_u$ and $\mathbf{F} \cdot \mathbf{e}_v$, with*

$$\mathbf{F} = (\nabla s \times \partial_s \mathbf{B} + \nabla u \times \partial_u \mathbf{B} + \nabla v \times \partial_v \mathbf{B}) \times \mathbf{B} - \mu_0 p' \nabla s. \quad (3)$$

The bracket is the curl of $\mathbf{B}$ in the coordinates $x^j = s, u, v$, since the Cartesian derivative of $\mathbf{B}$ is $\sum_j \nabla x^j \otimes \partial_j \mathbf{B}$. Every quantity in the statement is a formula in the reconstruction’s derivatives at the point, and with $\nabla p = p' \nabla s$ the outputs are the covariant components of $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$. `continuum_force` gives the same outputs through the Cartesian curl: for any $\mathbf{G} : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ and $P : \mathbf{R}^3 \rightarrow \mathbf{R}$ differentiable at $X(s_0, u_0, v_0)$ and equal to the reconstructed field and pressure along the three coordinate lines through it, they are $\mathbf{F} \cdot \mathbf{e}_i$ with $\mathbf{F} = (\nabla \times \mathbf{G}) \times \mathbf{G} - \mu_0 \nabla P$. `continuum_force_asym_coords` and `continuum_force_asym` are the two statements for a state without stellarator symmetry, and they rest on the identities at one point `Force.force_coords` and `Force.force_law`. At a node the solver’s rule is used instead: the angular outputs are the formulas above at the outer half point, the radial output is VMEC’s combination of the fields at the two half points, with centred differences for radial derivatives, and each of those fields is that of the series with VMEC’s half-grid values and slopes (`node_residual`, `node_residual_asym`).

VMEC's products. Every half-point quantity that is linear in the coefficients, R , Z and their derivatives, agrees with VMEC's term for term. The Jacobian and the metric elements are products, which VMEC forms from averages over the two nodes of products of node values, where the reconstruction multiplies the interpolated series. `VmecKernel.v` transcribes VMEC++'s kernels and proves the differences exactly. With r , z the even parts and $\hat{r}$, $\hat{z}$ the odd parts divided by $\sqrt{s}$ of the node values, and Δ the outer node's value less the inner one's, the reconstruction's factor $R_u Z_s - R_s Z_u$ and VMEC's τ satisfy

$$(R_u Z_s - R_s Z_u) - \tau = -\frac{1}{8}(\Delta \hat{r}_u \Delta \hat{z} - \Delta \hat{z}_u \Delta \hat{r}) - \frac{1}{8\sqrt{s_h}}(\Delta r_u \Delta \hat{z} - \Delta z_u \Delta \hat{r}) \quad (4)$$

(`vmec_jacobian_difference`), and each of g_{uu} , g_{uv} and g_{vv} differs from VMEC's by a sum of such products (`vmec_metric_difference`). Where every node difference in the identity is at most $K|h|$, the two Jacobian factors agree to within $\frac{1}{4}(Kh)^2(1 + s_h^{-1/2})$ (`vmec_jacobian_second_order`). The certified field is the reconstruction's, and VMEC's discrete field differs from it by these terms of second order in the radial step.

Identities. The representation builds some properties in, and for the reconstruction they are theorems rather than floating-point samples: it is solenoidal (`divergence_free`), the $(0,0)$ coefficient of λ drives nothing (`lambda_gauge`), which exhibits the poloidal relabelling as an exact kernel direction, and the residual has no component along the field (`residual_along_field`). The Euler–Lagrange expressions of VMEC's energy $W = \int (B^2/2 - \mu_0 p) \sqrt{g}$ at fixed λ , ι , flux and pressure are $-\sqrt{g}$ times the cylindrical components of the force (`energy_gradient_is_force`), so that the forces the solver drives to zero are the ideal-MHD force. Physical assumptions, such as second-order consistency of the discretization, are propositions in `Hypotheses.v` that the theorems using them take as hypotheses.

3 Certificates and their soundness

A certificate is a text file that a generator in `gen/` writes from a wout file. It carries every number of the file it uses as an integer mantissa and a power-of-two exponent, so no rounding happens between the file and the certificate, together with the bounds it claims; the extracted checker recomputes every enclosure from those numbers, and no bound the generator computes is trusted.

Expressions and intervals. `Expr.v` gives an expression two meanings: a value over extended reals, in which division by zero and the square root of a negative number give `Xnan`, and a value over CoqInterval's floating-point intervals. `ieval_correct` states that the interval contains the extended-real value, so a bounded output interval proves both that the value is a real number and that it lies in the bounds. Because the embedding is deep, one tree carries both meanings, `Deriv.v` differentiates it symbolically with a chain rule proven in the extended-real sense, and subexpressions that the residual's components share are bound once to environment slots.

Verdicts. A point certificate bounds the residual at points (`check_cert_correct`). A cell certificate bounds a component over a rectangle of two varied slots, two angles or an angle and the radius, by its value at the centre plus enclosures of its derivatives over the rectangle times the half-widths (`check_ccert_correct`); reversed, the same data bounds a component away from zero. That bound is loose on coarse cells, since interval enclosures of a derivative over a whole

cell overestimate: the eight-cell regression certificate of the Solov’ev equilibrium in the repository bounds residuals no larger than 1.5×10^{-7} at the cell centres by 29 to 2200 over the cells. `Cover2.v` decides, column by column, that the cells of a verdict cover the rectangle they span (`cells_cover_rect`) or every line they lie on (`cells_cover_lines`). A spectral solution balances forces mode by mode, so the quantity it makes small is the discrete Fourier harmonic of the residual at a retained mode, which `Project.v` encloses (`harm_encloses`); `Harmonic.v` integrates exactly the components whose trigonometric degree lies below an equispaced grid’s point counts, and `Integrate.v` encloses other flux-surface integrals by the midpoint rule over a lattice of cells that the checker lays out itself, with the error read off its own enclosures of the second derivatives over each cell (`integ2_correct`, `integ1_correct`). `MercierRun.v` assembles DShear, DCurr, DWell, DGeod and DMerc as VMEC’s `mercier.f90` defines them [10, 11] from integrals over the whole angular torus that the same run computes: the lattice reaches past each period of 2π , the period ends inside its last cells, and the two strips it adds are bounded through the integrand’s enclosures over those cells (`integ2_torus_correct`). A verdict is then a theorem about the criterion of the reconstruction rather than about its ingredients (`merc_run_correct`). In `Box.v` every input slot carries a half-width, so one evaluation covers every field whose coefficients lie in a box.

Discrete equilibria. `Newton.v` proves that a passing interval Newton test in Krawczyk form [12, 13] on any system of expressions gives exactly one zero in its box: $x - AF(x)$ maps the box into itself as a contraction, and an approximate inverse A makes its fixed point a zero of F . `Colloc.v` collocates the residual over a band of surfaces as such a system, with the surface coefficients as unknowns (`colloc_correct`), and `Wide.v` evaluates the system’s outputs at the centre of the box in floating point of any precision.

Trust base. The kernel checks the proofs; what runs is the OCaml extraction of the checking functions, linked with a driver that parses certificates into the extracted data types, divides the checking among processes and prints verdicts. The trusted components are the Rocq kernel and extraction, the OCaml toolchain, the implementation of Rocq’s primitive 63-bit integers and binary64 floats, the driver’s parser and its division of the work, and the correspondence between a certificate’s numbers and the output or formulas they are taken from, which `gen/verify_cert.py` checks for a wout file. The classical constructions behind Rocq’s real numbers appear only in specifications, and extraction directives replace them by values that fail when consulted. For the theorems named here, `Print Assumptions` reports the two axioms of the standard library’s classical Dedekind reals, excluded middle, functional extensionality and the specifications of the primitive operations. The build pins Rocq 9.1.1 with its standard library 9.2.0, CoqInterval 4.11.4, Flocq 4.2.2, Coquelicot 3.4.5, MathComp 2.4.0 and OCaml 4.14.2.

4 Results

Table 1 collects the certified results with the theorem each rests on. The manufactured problems are Stellarocq’s collocated systems for VMEC++’s manufactured mappings, and every other case is a VMEC++ run or a file from its test data. A number called certified is an enclosure that a passing verdict proves, rounded outward; floating-point numbers are named as such.

The order of the half-grid rule. One, two and three surfaces of the Solov’ev equilibrium [14] each have exactly one zero of the collocated residual in a box shaped to the coefficients, of radius 5.7×10^{-10} to 6.3×10^{-9} , around a centre that Newton’s method moved 2.2×10^{-6} to 3.5×10^{-6}

Case	Certified quantity	Enclosure
manufactured, 3D	$\max x_h - x^* $, 9/17/33/65 surfaces <code>colloc_correct</code>	$2.73432936829 \times 10^{-6}$, $7.16912170772 \times 10^{-7}$, $1.83032585742 \times 10^{-7}$, $4.6074819334 \times 10^{-8}$
manufactured, asymmetric	$\max x_h - x^* $, 9/17/33 surfaces <code>colloc_correct</code>	$2.69073057043 \times 10^{-5}$, $7.20667053290 \times 10^{-6}$, $1.86266877703 \times 10^{-6}$
Solov'ev, 1–3 surfaces	zero of the collocated residual <code>colloc_correct</code>	exactly one, in boxes of radius 5.7×10^{-10} to 6.3×10^{-9}
rippled tokamak, $\iota = 1/2$ at $s = 0.625$	(2, 1) harmonic of r_s at the converged state and over a box of relative width 10^{-14} , 65/129/257 surfaces <code>dharm_correct</code>	[8.34, 8.46], [8.67, 9.08], $[6.19, 7.77] \times 10^{-5} \text{ T}^2$; over the box [8.29, 8.51], [8.45, 9.29], $[5.32, 8.64] \times 10^{-5} \text{ T}^2$
CTH-like, free boundary	largest jump of $\mu_0 p + B^2/2$ against $B_{\text{vac}}^2/2$ on NESTOR's grid, 25/385 surfaces <code>check_bpcert_correct</code>	$[7.556, 7.561] \times 10^{-4} \text{ T}^2$, $[4.9647, 4.9673] \times 10^{-4} \text{ T}^2$, for every vacuum pressure between NESTOR's and BIEST's
CTH-like, free boundary	the same jump over the whole boundary, 25/385 surfaces <code>check_btcert_correct</code>	at most $8.47 \times 10^{-3} \text{ T}^2$, at most $1.051 \times 10^{-2} \text{ T}^2$
up_down_asym, $s = 0.125$, 17 surfaces	Mercier current term and DMerc <code>merc_run_correct</code>	$[-3.724, -3.716] \times 10^{-3}$ and $[-1.371, -1.205] \times 10^{-3}$, where the file gives -6.1×10^{-4} and $+1.8 \times 10^{-3}$ in floating point

Table 1: Certified results on the manufactured problems and on VMEC++ equilibria, with the soundness theorem behind each.

from the file's coefficients. `HalfGrid.v` proves the node rule second-order consistent with the residual of the continuously reconstructed field (`node_consistent`) and derives the second-order bound that `discretization_is_consistent` states from consistency and a stability hypothesis (`lax_second_order`); near the axis, where the innermost interval is read linearly, the rule is first order in the radial derivatives. Manufactured solutions [15] measure the order. A manufactured mapping on the annulus $1/4 < s < 1$ is not an equilibrium; its continuum residual at the collocation points is a source f , and the zero x_h of $r - f$ over the interior nodes, with λ given so that the problem sits on the quotient by the poloidal gauge, is the discrete solution. Every datum of the problem, the coefficients it holds fixed, λ , the transform and the source, is the mapping's own value carried to 120 bits. On systems of up to 423 unknowns the interval Newton test puts exactly one zero in a box a few units of a 58-bit mantissa wide in each unknown, around a centre refined on outputs read at 128 bits, so the error $\max |x_h - x^*|$ is enclosed to within 5×10^{-20} , a relative 10^{-12} , with no solver tolerance in it; the same problem with its data rounded to binary64 has an error that differs in the eighth digit. It falls by 3.81, 3.92 and 3.97 per doubling for a three-dimensional mapping and by 3.73 and 3.87 for one without stellarator symmetry (Figure 1a), and its enclosures are identical at five times the pressure, which enters the radial force only as a flux function that the rule and its source share. Two limits bound what this measures. The collocated system is the half-grid residual of the reconstruction at collocation points, not VMEC's own discrete equations, which come from the variation of the energy with spectral condensation and a constraint on the $m = 1$ modes and form the Jacobian and metric from VMEC's products. And the annulus leaves out the

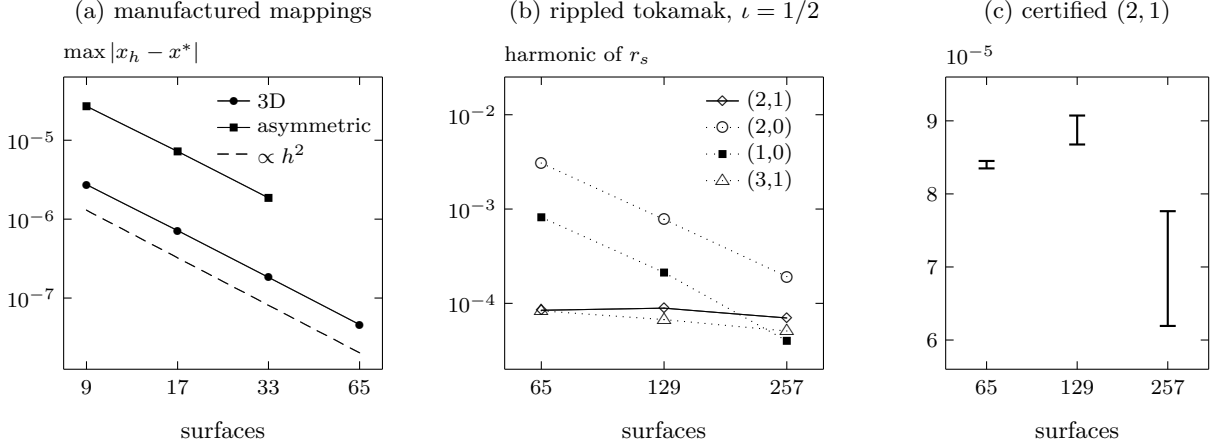


Figure 1: (a) The certified error of the half-grid scheme on the manufactured mappings; the dashed line falls as h^2 . (b) Harmonics of the radial force r_s on the surface $\iota = 1/2$ of the rippled tokamak: the certified resonant (2, 1) harmonic and the floating-point (2, 0), (1, 0) and (3, 1) harmonics of the same reconstruction. The (2, 0) and (1, 0) harmonics fall 3.8 to 5.3-fold per doubling, the resonant harmonic and its (3, 1) sideband at most 1.5-fold. (c) The certified enclosures of the (2, 1) harmonic on a linear scale.

axis, where the rule is first order. The first-order radial convergence that [16] report for VMEC is of a different quantity, the force error of VMEC’s own solutions averaged over $0.1 \leq s \leq 0.99$ of a discrete problem that contains the axis.

A forced current sheet. On VMEC++’s circular tokamak ($R_0 = 6$, $a = 2$) with $\iota = 0.9 - 0.64s$, which is $1/2$ at $s = 0.625$, and a (1, 1) boundary ripple of amplitude 0.03, toroidal coupling drives a (2, 1) harmonic of the radial force on the rational surface. Its certified value, the rectangle rule for the integral of $r_s \cos(2u - v)$ on a 64×32 grid of angles, in T^2 like r_s , against $1.2 \times 10^3 \text{ T}^2$ for the integral of B^2 over the angles there, enclosed by `dharm_correct` at the converged state, lies between 6.1×10^{-5} and 9.1×10^{-5} at 65, 129 and 257 surfaces (Figure 1c), and between 5.3×10^{-5} and 9.3×10^{-5} for every state within a relative 10^{-14} of the converged coefficients; it is linear in the ripple. From 65 to 129 surfaces it rises by a factor between 1.02 and 1.09, and from 129 to 257 it falls by a factor between 1.11 and 1.47, where the axisymmetric harmonics (2, 0) and (1, 0) of the same reconstruction fall 3.8 to 5.3-fold per doubling in floating point and the (3, 1) sideband 1.2 to 1.3-fold (Figure 1b). The same ripple on a transform profile with no low-order rational surface nearby leaves nothing behind. This is the response of a nested-surface field at a rational surface [17, 18, 19], whose ideal limit is a current sheet and which [20] studied in VMEC; three-dimensional equilibria with current sheets and stepped pressure exist [21, 22]. Two limits apply. Three resolutions show a harmonic falling far more slowly than the scheme’s order; they do not decide whether it tends to a nonzero limit, and the enclosures widen about fourfold per doubling, from 1.2 to 22 per cent of the value, as second radial differences amplify the rounding of the binary64 interval arithmetic by h^{-2} . And the box is a box around the converged state: near a rational surface the linearized operator is nearly singular, so the box need not contain the exact discrete solution, and the collocated Newton test that would certify one passes on that surface at 33 surfaces and not at 65, where the Jacobian’s condition number is five times larger. In SPEC [23], two Beltrami volumes separated by an ideal interface at the rational surface carry the (2, 1) sheet as a jump of 1.47×10^{-5} in the covariant poloidal field at ripple 0.03, as in the equilibria of [24],

and with the rational surface inside a relaxed volume SPEC's field-line tracing finds a $(2, 1)$ island whose width grows from 0.107 to 0.170 in the volume's radial coordinate as the ripple goes from 0.0075 to 0.03, as the ripple to the power 0.33.

Free-boundary balance. At a free-boundary equilibrium $\mu_0 p + B^2/2$ is continuous across the plasma boundary, where outside it is the pressure of the full vacuum field. VMEC++ computes that field with NESTOR [25, 2], and BIEST's virtual-casing field [26, 27] gives it independently. The certificate sweeps the vacuum pressure over the segment between the two answers, $P_{\text{vac}} = P_{\text{mid}} + t \delta P$ with $t \in [-1, 1]$, and takes the plasma side as VMEC's extrapolation of the two outermost half points to $s = 1$. Neither answer is certified, so every statement below holds for the vacuum fields of that segment and for the true one only if it lies there. On the CTH-like test case the largest jump on NESTOR's grid is enclosed in $[7.556, 7.561]$, $[7.201, 7.206]$, $[6.433, 6.437]$, $[5.636, 5.640]$ and $[4.9647, 4.9673] \times 10^{-4} \text{ T}^2$ at 25, 49, 97, 193 and 385 surfaces, for every t , where $B^2/2$ averages 0.18 T^2 on the boundary. Each enclosure lies below the one before it, so the jump falls under radial refinement, but only by 1.05 to 1.14 per doubling. VMEC imposes the balance through the Fourier harmonics it retains, and in floating point the jump on NESTOR's grid lies almost wholly beyond them: at 49 surfaces its part in the retained harmonics is at most 2.4×10^{-5} and the rest 7.2×10^{-4} . The retained part falls 3.3 and 1.9-fold per doubling from 25 to 97 surfaces, while with NESTOR's poloidal grid fixed, retaining 5 to 12 poloidal harmonics moves the jump between 4.0×10^{-4} and 1.5×10^{-3} without reducing it. Covered over the whole boundary by 512×128 cells per field period, with the vacuum side the trigonometric interpolant of each answer's grid, the jump is at most 8.47×10^{-3} at 25 surfaces and 1.051×10^{-2} at 385, 11 and 21 times the bound on the grid. Between grid points the bound is set by that interpolant, whose angular resolution radial refinement leaves as it is, so it does not fall.

The Mercier criterion at coarse resolution. At $s = 0.125$ of `up_down_asym` on 17 surfaces, with 2048 poloidal cells, the certified Mercier current term lies in $[-3.724, -3.716] \times 10^{-3}$ where the file's difference quotients give -6.1×10^{-4} , and the certified DMerc lies in $[-1.371, -1.205] \times 10^{-3}$ where the file's is $+1.8 \times 10^{-3}$: the reconstruction's criterion is certified unstable where the file's is stable. From 33 to 257 surfaces both current terms tend to zero with the file's sign, the certified one from $[-1.092, -1.083] \times 10^{-3}$ to $[-3.7, -1.8] \times 10^{-5}$, and the certified DMerc is positive at each. At 17 surfaces the sign of DMerc depends on how the radial derivatives are taken, and a certified value at a free radius is that of the Hermite reconstruction, whose radial interpolation is a choice.

Use in VMEC++. The certified values enter a branch of VMEC++ [28] as regression tests whose tolerances are the enclosures: the manufactured-solution errors, the resonant floor with the fall of the axisymmetric harmonics, and the free-boundary jump, which reads NESTOR's boundary vacuum field through an option proposed upstream [29]. A finite difference of VMEC++'s energy equals its unpreconditioned forces contracted with the perturbation, the discrete form of `energy_gradient_is_force`; VMEC++ now tests this [30], and that the energy Hessian is symmetric and DGeod never positive [31]. Differential tests of VMEC++ for invariances the reconstruction satisfies exactly, three of which VMEC++ now runs [32], also exposed an orientation test that the poloidal rotation VMEC applies to an asymmetric boundary can defeat: the test multiplies two $m = 1$ coefficients whose product the rotation can drive to zero, where the invariant is the $m = 1$ determinant $r^{cc}z^{sc} - r^{sc}z^{cc}$. Relabelling the `up_down_asym_current` boundary by $\theta \rightarrow \pi - \theta$ sends educational_VMEC to its iteration cap at $f_{sq\bar{r}} = 2.07 \times 10^{-2}$ instead of converging to 5.07×10^{-12} . It was reported as `vmecpp#788` [33], after which VMEC++ tests the determinant [34], and as

5 Related work

Rigorous numerics proves statements about dynamical systems and partial differential equations with interval arithmetic: the existence of the Lorenz attractor [36], the CAPD library for ordinary differential equations [37], KAM theory applied by computer [38], among other systems to magnetic field lines [39], blow-up of the Boussinesq and axisymmetric Euler equations [40], and elliptic and parabolic problems [41]. These proofs run in libraries outside a proof assistant. Inside one, verified ODE solvers up to the Lorenz attractor [42, 43], the discretization and floating-point program of the one-dimensional wave equation [44], the Lax–Milgram theorem [45] and the certificates of the Kepler conjecture [46] come closest; in Lean, high-energy physics is being formalized [47], the characterization of the equilibria of the Vlasov–Maxwell–Landau system has been [48], and smooth magnetohydrostatic equilibria with nested pressure surfaces and only a discrete rotational symmetry, counterexamples to Grad’s conjecture [17], come with a Lean certification [49]. In fusion, formal methods have verified a plasma vertical-position control algorithm [50] and properties of neural surrogates of the Grad–Shafranov equation [51]. Stellarator equilibrium codes are verified by convergence studies, manufactured solutions and comparison between codes, VMEC against SPEC [23, 24, 20] and against DESC [52, 16]. We know of no prior work that states the output of a stellarator equilibrium code as a machine-checked theorem about the field it defines.

6 Conclusion

Stellarocq gives the output of an equilibrium code a meaning a proof assistant can check: bounds, harmonics, stability terms and the existence of discrete solutions, each a theorem about the field reconstructed from the file’s exact numbers, whose residual is proven to be the ideal-MHD force. It does not reach the step from a small residual of the reconstruction to a nearby equilibrium of the continuum problem, which needs a bound on the inverse of the linearized force operator that the development states as a hypothesis and does not prove. Its verdicts enter VMEC++ as regression tests whose tolerances are the certified enclosures. Landreman’s explicit three-dimensional equilibria with nested surfaces [53] would let the same certificates measure a solver’s error against a true equilibrium instead of a manufactured one; a floating-point comparison of VMEC++ with them is proposed upstream [54].

Availability. The development, the certificate generators and the script that writes Figure 1 are at <https://github.com/CharlesCNorton/stellarocq>, commit `ed255b9`; `make all` builds the proofs and the checker, `make audit` prints the assumptions of every theorem named here, the release `checker-1ce73cc` carries the same checker linked statically for `x86_64` Linux, and each generator’s documentation gives the commands of its result. The VMEC++ tests that carry the certified values are on the branch of [28].

References

- [1] S. P. Hirshman and J. C. Whitson, Steepest-descent moment method for three-dimensional magnetohydrodynamic equilibria, *Phys. Fluids* **26**, 3553 (1983).
- [2] S. P. Hirshman, W. I. van Rij and P. Merkel, Three-dimensional free boundary calculations using a spectral Green’s function method, *Comput. Phys. Commun.* **43**, 143 (1986).

- [3] J. Schilling, The numerics of VMEC++, arXiv:2502.04374 (2025).
- [4] J. Schilling, E. Guiraud and V. Siska, VMEC++, software, doi:10.5281/zenodo.14800158.
- [5] The Rocq Development Team, The Rocq Prover, <https://rocq-prover.org>.
- [6] P. Letouzey, A new extraction for Coq, in *Types for Proofs and Programs (TYPES 2002)*, LNCS 2646, 200 (2003).
- [7] S. Boldo, C. Lelay and G. Melquiond, Coquelicot: A user-friendly library of real analysis for Coq, *Math. Comput. Sci.* **9**, 41 (2015).
- [8] S. Boldo and G. Melquiond, Flocq: A unified library for proving floating-point algorithms in Coq, in *20th IEEE Symposium on Computer Arithmetic*, 243 (2011).
- [9] G. Melquiond, Proving bounds on real-valued functions with computations, in *IJCAR 2008*, LNCS 5195, 2 (2008).
- [10] C. Mercier, A necessary condition for hydromagnetic stability of plasma with axial symmetry, *Nucl. Fusion* **1**, 47 (1960).
- [11] F. Bauer, O. Betancourt and P. Garabedian, *Magnetohydrodynamic Equilibrium and Stability of Stellarators* (Springer, New York, 1984).
- [12] R. Krawczyk, Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken, *Computing* **4**, 187 (1969).
- [13] R. E. Moore, R. B. Kearfott and M. J. Cloud, *Introduction to Interval Analysis* (SIAM, Philadelphia, 2009).
- [14] L. S. Solov'ev, The theory of hydromagnetic stability of toroidal plasma configurations, *Sov. Phys. JETP* **26**, 400 (1968).
- [15] P. J. Roache, Code verification by the method of manufactured solutions, *J. Fluids Eng.* **124**, 4 (2002).
- [16] D. Panici, R. Conlin, D. W. Dudt, K. Unalmis and E. Kolemen, The DESC stellarator code suite. Part 1. Quick and accurate equilibria computations, *J. Plasma Phys.* **89**, 955890303 (2023).
- [17] H. Grad, Toroidal containment of a plasma, *Phys. Fluids* **10**, 137 (1967).
- [18] T. S. Hahm and R. M. Kulsrud, Forced magnetic reconnection, *Phys. Fluids* **28**, 2412 (1985).
- [19] J. Loizu, S. R. Hudson, A. Bhattacharjee and P. Helander, Magnetic islands and singular currents at rational surfaces in three-dimensional magnetohydrodynamic equilibria, *Phys. Plasmas* **22**, 022501 (2015).
- [20] S. A. Lazerson, J. Loizu, S. Hirshman and S. R. Hudson, Verification of the ideal magnetohydrodynamic response at rational surfaces in the VMEC code, *Phys. Plasmas* **23**, 012507 (2016).
- [21] O. P. Bruno and P. Laurence, Existence of three-dimensional toroidal MHD equilibria with nonconstant pressure, *Commun. Pure Appl. Math.* **49**, 717 (1996).
- [22] A. Enciso, A. Luque and D. Peralta-Salas, MHD equilibria with nonconstant pressure in nondegenerate toroidal domains, *J. Eur. Math. Soc.* **27**, 2251 (2025).
- [23] S. R. Hudson, R. L. Dewar, G. Dennis, M. J. Hole, M. McGann, G. von Nessi and S. Lazerson, Computation of multi-region relaxed magnetohydrodynamic equilibria, *Phys. Plasmas* **19**, 112502 (2012).
- [24] J. Loizu, S. R. Hudson, A. Bhattacharjee, S. Lazerson and P. Helander, Existence of three-dimensional ideal-magnetohydrodynamic equilibria with current sheets, *Phys. Plasmas* **22**, 090704 (2015).
- [25] P. Merkel, An integral equation technique for the exterior and interior Neumann problem in toroidal regions, *J. Comput. Phys.* **66**, 83 (1986).

- [26] D. Malhotra, A. J. Cerfon, L.-M. Imbert-Gérard and M. O’Neil, Taylor states in stellarators: A fast high-order boundary integral solver, *J. Comput. Phys.* **397**, 108791 (2019).
- [27] D. Malhotra, A. J. Cerfon, M. O’Neil and E. Toler, Efficient high-order singular quadrature schemes in magnetic fusion, *Plasma Phys. Control. Fusion* **62**, 024004 (2020).
- [28] VMEC++ with Stellarocq’s certified values as regression tests, <https://github.com/CharlesCNorton/vmecpp/tree/stellarocq-certificates>.
- [29] Return NESTOR’s boundary vacuum field with `return_vacuum_field`, <https://github.com/proximafusion/vmecpp/pull/903>.
- [30] Test that the unpreconditioned force is the gradient of the MHD energy, <https://github.com/proximafusion/vmecpp/pull/905>.
- [31] Test that the energy Hessian is symmetric and DGeod is never positive, <https://github.com/proximafusion/vmecpp/pull/904>.
- [32] Test three invariances of the solver, <https://github.com/proximafusion/vmecpp/pull/875>.
- [33] The Jacobian orientation check is not invariant under the poloidal rotation applied to an asymmetric boundary, <https://github.com/proximafusion/vmecpp/issues/788>.
- [34] Test the Jacobian orientation with a quantity the lasym rotation preserves, <https://github.com/proximafusion/vmecpp/pull/789>.
- [35] The Jacobian orientation test is not invariant under the poloidal rotation applied to an asymmetric boundary, https://github.com/jonathanschilling/educational_VMEC/issues/27.
- [36] W. Tucker, The Lorenz attractor exists, *C. R. Acad. Sci. Paris Sér. I Math.* **328**, 1197 (1999).
- [37] T. Kapela, M. Mrozek, D. Wilczak and P. Zgliczyński, CAPD::DynSys: a flexible C++ toolbox for rigorous numerical analysis of dynamical systems, *Commun. Nonlinear Sci. Numer. Simul.* **101**, 105578 (2021).
- [38] J.-Ll. Figueras, A. Haro and A. Luque, Rigorous computer-assisted application of KAM theory: a modern approach, *Found. Comput. Math.* **17**, 1123 (2017).
- [39] L. Valvo and U. Locatelli, Hamiltonian control of magnetic field lines: computer assisted results proving the existence of KAM barriers, arXiv:2101.07785 (2021).
- [40] J. Chen and T. Y. Hou, Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data II: Rigorous numerics, arXiv:2305.05660 (2023).
- [41] M. T. Nakao, M. Plum and Y. Watanabe, *Numerical Verification Methods and Computer-Assisted Proofs for Partial Differential Equations* (Springer, Singapore, 2019).
- [42] F. Immler and J. Hölzl, Numerical analysis of ordinary differential equations in Isabelle/HOL, in *Interactive Theorem Proving (ITP 2012)*, LNCS 7406, 377 (2012).
- [43] F. Immler, A verified ODE solver and the Lorenz attractor, *J. Autom. Reason.* **61**, 73 (2018).
- [44] S. Boldo, F. Clément, J.-C. Filliâtre, M. Mayero, G. Melquiond and P. Weis, Wave equation numerical resolution: A comprehensive mechanized proof of a C program, *J. Autom. Reason.* **50**, 423 (2013).
- [45] S. Boldo, F. Clément, F. Faissole, V. Martin and M. Mayero, A Coq formal proof of the Lax–Milgram theorem, in *6th ACM SIGPLAN Conference on Certified Programs and Proofs (CPP 2017)*, 79 (2017).
- [46] T. Hales et al., A formal proof of the Kepler conjecture, *Forum Math. Pi* **5**, e2 (2017).
- [47] J. Tooby-Smith, HepLean: Digitalising high energy physics, *Comput. Phys. Commun.* **308**, 109457 (2025).
- [48] V. Ilin, Semi-autonomous formalization of the Vlasov–Maxwell–Landau equilibrium, arXiv:2603.15929 (2026).

- [49] J. Gómez-Serrano, L. Liehr and M. A. Taylor, Counterexamples to Grad’s conjecture, arXiv:2609.24739 (2026).
- [50] M. Wu, J. Rosenberg and N. Fulton, A formally verified plasma vertical position control algorithm, in *Formal Methods for Industrial Critical Systems (FMICS 2020)*, LNCS 12327, 170 (2020).
- [51] F. N. Rizqan, M. Hole and C. Gretton, Evaluation and verification of physics-informed neural models of the Grad–Shafranov equation, arXiv:2504.21155 (2025).
- [52] D. W. Dudt and E. Kolemen, DESC: A stellarator equilibrium solver, *Phys. Plasmas* **27**, 102513 (2020).
- [53] M. Landreman, Analytic toroidal 3D MHD equilibria and steady Euler flows with invariant surfaces, arXiv:2609.26742 (2026).
- [54] Check VMEC++ against exact three-dimensional MHD equilibria in CI, <https://github.com/proximafusion/vmecpp/pull/912>.